Multimodal decision models for camera-trap bird monitoring

Evaluation architecture · one shared typed-question request per frame · video-disjoint measurement

implemented request path
experimental extension

A / FRAME-LEVEL INFERENCE CONTRACT

1 SOURCE

Video archive



Camera-trap video files

H55 + additional site datasets

PARTITION KEY

video_id

All frames from a clip stay in the same train / val / test partition to limit leakage.

`V = {v_1, ..., v_n}`

2 TEMPORAL SAMPLING

Decode & sample

`v_i -> {I_i,t}`

Stride delta_t; retain source index t



Selected frames preserve order; frame map links I_i,t back to v_i.

PER-FRAME INPUT

```
I_i,t : RGB image
s_i,t : neutral metadata
```

No annotation-derived text is placed in the request.

3 REQUEST ASSEMBLY

Typed question map

`Q = {q_1, q_2, ..., q_K}`

noul	bird_present	2 outcomes
score	bird_count	4 levels
choice	largest_bird_cell	10 options
choice	bird_behavior*	9 options

WIRE ENVELOPE

```
{ state: s_i,t,
images: [I_i,t],
questions: Q }
```

*Behavior remains exploratory until labeled.

Q + media

4 LOCAL INFERENCE

Backend adapters

Common contract; implementations vary

- **Clef / Clef-Flash** — Ollama System One · local
- **Liquid d1** — Local runtime / adapter dependent
- **Jev-Omni** — MLX / CUDA; current adapter calls per question
- **OneJev** — Local compatible server

CLIENT-SIDE GUARANTEE

1 request / frame, full Q included

Not a claim of one internal model pass

`POST /v1/systemone`

5 NORMALIZE & MEASURE

Typed outputs

`P_i,t = {p(q_j | I_i,t)}`

Per-question probability vectors

```
noul: [p_false, p_true]
score: [p_0, ..., p_L−1]
choice: [p_1, ..., p_K]
```

EVALUATE ONLY WITH GOLD

Presence · count · location

Behavior metrics require reviewed labels

```
metrics.json + decisions.json
answers · errors · usage · provenance
```

Gold: presence · boxes · behavior
Score available labels; split by video.

B / EXTENSIONS FOR FOLLOW-UP EXPERIMENTS · NOT PART OF THE CURRENT PER-FRAME BASELINE

Temporal context / decision reuse

`(I_t, p_t−1, c_t−1) -> gate -> p_t`

new frame I_t → scene-change / max-age gate → infer or reuse cached decision

Compare with every-frame inference; measure recall loss and latency saved. Never forward-fill scores without validation.

Calibration / model adaptation

gold train videos (labels + media) → temperature / bias or head / LoRA → adapted θ′ candidate

Select on validation videos; freeze the test partition.

Retain frozen-backbone results as the primary baseline.

Optional box-output pathway

box-capable model (e.g., PBD system) → `(x_1,y_1, x_2,y_2)` → IoU / AP vs. gold boxes

The current 3 × 3 cell answer is coarse localization,

not a predicted box. PBD concerns coordinate decoding.